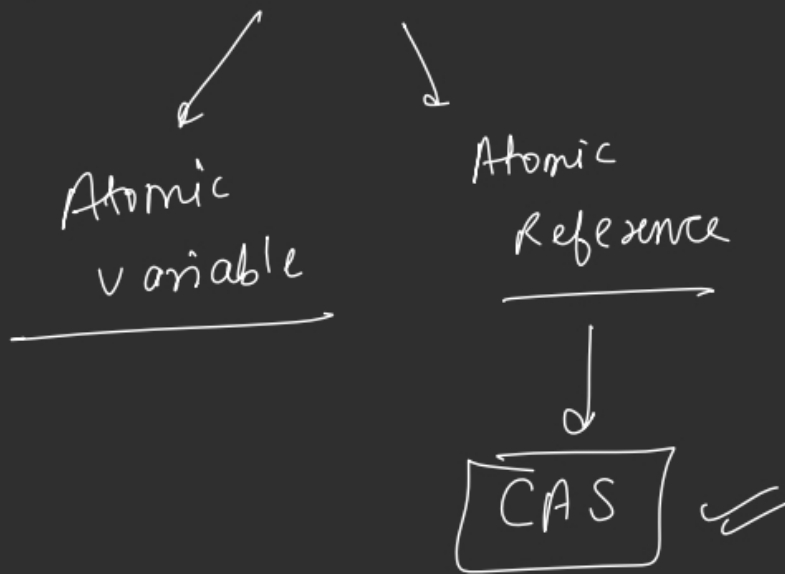
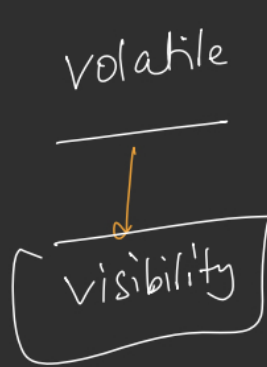


# lock free Concurrency

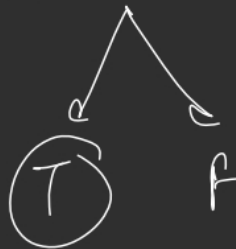


volatile

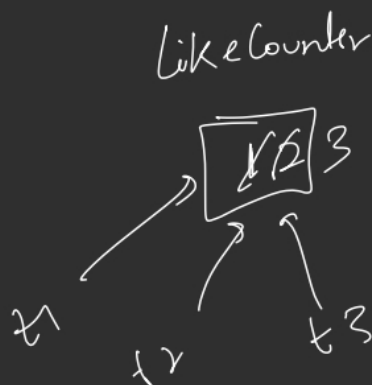
atomic



seat. Compare And Set (expectedValue, newValue)



Lock-free Re-try loop



→ Re-calculate ✓✓

[AtomicReferenceArray]

```
int[] arr = new int[10];
```

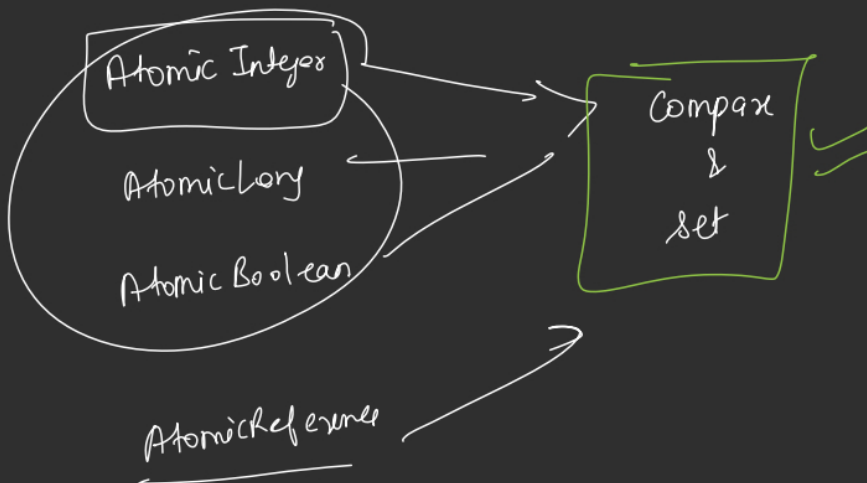
AtomicReferenceArray<String> arr =

new AtomicReferenceArray<>(5);

arr.set(0, "Aditya");

arr.set(1, "Rohit");

↓  
arr.compareAndSet(2, "Rohan", "Sohan");



Compare & Set + re-by logic

Compare And Swap

read → modify → write

read  $\rightarrow$  modify  $\rightarrow$  write

Lock  
↓↓↓↓↓  
t1

CAS  
↓↓↓  
try  
only one  
succeeds.

CAS  
↓  
ABA Problem

CAS

initial = 'A' 'B' 'A'

'A'  
version

| t1       | t2         |
|----------|------------|
| t1 = 'A' |            |
|          | t2 = 'A' ✓ |
|          | t2 = B     |
|          | t2 = B A   |
| t1       |            |

like Counter = ~~1~~  
2

Versioning  $\rightarrow$  (value, version)

